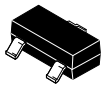
Digital FET, P-Channel

FDV304P, FDV304P-F169



SOT−23−3
CASE 318−08

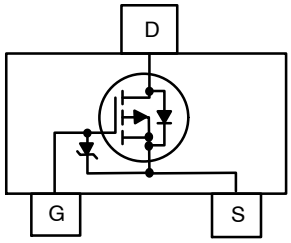
General Description

This P−Channel enhancement mode field effect transistors is produced using **onsemi**'s proprietary, high cell density, DMOS technology. This very high density process is tailored to minimize on−state resistance at low gate drive conditions. This device is designed especially for application in battery power applications such as notebook computers and cellular phones. This device has excellent on−state resistance even at gate drive voltages as low as 2.5 V.

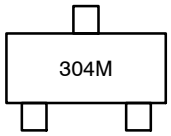
Features

- −25 V, −0.46 A Continuous, −1.5 A Peak
  - $R_{DS(on)}$ = 1.1 Ω @ V_{GS} = −4.5 V
  - $R_{DS(on)}$ = 1.5 Ω @ V_{GS} = −2.7 V
- Very Low Level Gate Drive Requirements Allowing Direct Operation in 3 V Circuits. $V_{GS(th)}$ < 1.5 V
- Gate−Source Zener for ESD Ruggedness. > 6 kV Human Body Model
- Compact Industry Standard SOT−23 Surface Mount Package
- This Device is Pb−Free and Halide Free

ELECTRICAL CONNECTION



MARKING DIAGRAM



304 = Specific Device Code
M = Date Code

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Value	Unit
V_{DSS}	Drain−Source Voltage	−25	V
V_{GSS}	Gate−Source Voltage	−8	V
I_D	Drain Current − Continuous	−0.46	A
	Drain Current − Pulsed	−1.5	
P_D	Maximum Power Dissipation	0.35	W
T_J, T_{STG}	Operating and Storage Temperature Range	−55 to 150	°C
ESD	Electrostatic Discharge Rating MIL−STD−883D Human Body Model (100 pF/1500 Ω)	6.0	kV

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction−to−Ambient	357	°C/W

ORDERING INFORMATION

Device	Package	Shipping†
FDV304P	SOT−23−3 (Pb−Free, Halide−Free)	3000 / Tape & Reel
FDV304P−F169	SOT−23−3 (Pb−Free, Halide−Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
BV_{DSS}	Drain−Source Breakdown Voltage	V_{GS} = 0 V, I_D = −250 μA	−25	−	−	V
$\Delta BV_{DSS}/\Delta T_J$	Breakdown Voltage Temp. Coefficient	I_D = −250 μA, Referenced to 25°C	−	−22	−	mV/°C
I_{DSS}	Zero Gate Voltage Drain Current	V_{DS} = −20 V, V_{GS} = 0 V	−	−	−1	μA
		V_{DS} = −20 V, V_{GS} = 0 V, T_J = 55°C	−	−	−10	
I_{GSS}	Gate−Body Leakage Current	V_{GS} = −8 V, V_{DS} = 0 V	−	−	−100	nA
ON CHARACTERISTICS (Note 1)						
$\Delta V_{GS(th)}/\Delta T_J$	Gate Threshold Voltage Temp. Coefficient	I_D = −250 μA, Referenced to 25°C	−	2.1	−	mV/°C
$V_{GS(th)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = −250 μA	−0.65	−0.86	−1.5	V
$R_{DS(on)}$	Static Drain−Source On−Resistance	V_{GS} = −2.7 V, I_D = −0.25 A	−	1.22	1.5	Ω
		V_{GS} = −4.5 V, I_D = −0.5 A	−	0.87	1.1	
		V_{GS} = −4.5 V, I_D = −0.5 A, T_J = 125°C	−	1.21	2	
$I_{D(on)}$	On−State Drain Current	V_{GS} = −2.7 V, V_{DS} = −5 V	−0.5	−	−	A
		V_{GS} = −4.5 V, V_{DS} = −5 V	−1	−	−	
g_{FS}	Forward Transconductance	V_{DS} = −5 V, I_D = −0.5 A	−	0.8	−	S
DYNAMIC CHARACTERISTICS						
C_{iss}	Input Capacitance	V_{DS} = −10 V, V_{GS} = 0 V, f = 1.0 MHz	−	63	−	pF
C_{oss}	Output Capacitance		−	34	−	
C_{rss}	Reverse Transfer Capacitance		−	10	−	
SWITCHING CHARACTERISTICS (Note 1)						
$t_{D(on)}$	Turn−On Delay Time	V_{DD} = −6 V, I_D = −0.5 A, V_{GS} = −4.5 V, R_{GEN} = 50 Ω	−	7	20	ns
t_r	Turn−On Rise Time		−	8	20	
$t_{D(off)}$	Turn−Off Delay Time		−	55	110	
t_f	Turn−Off Fall Time		−	35	70	
Q_g	Total Gate Charge	V_{DS} = −5 V, I_D = −0.25 A, V_{GS} = −4.5 V	−	1.1	1.5	nC
Q_{gs}	Gate−Source Charge		−	0.32	−	
Q_{gd}	Gate−Drain Charge		−	0.25	−	
DRAIN−SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS						
I_S	Maximum Continuous Drain−Source Diode Forward Current		−	−	−0.5	A
V_{SD}	Drain−Source Diode Forward Voltage	V_{GS} = 0 V, I_S = −0.5 A (Note 1)	−	−0.89	−1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

TYPICAL CHARACTERISTICS

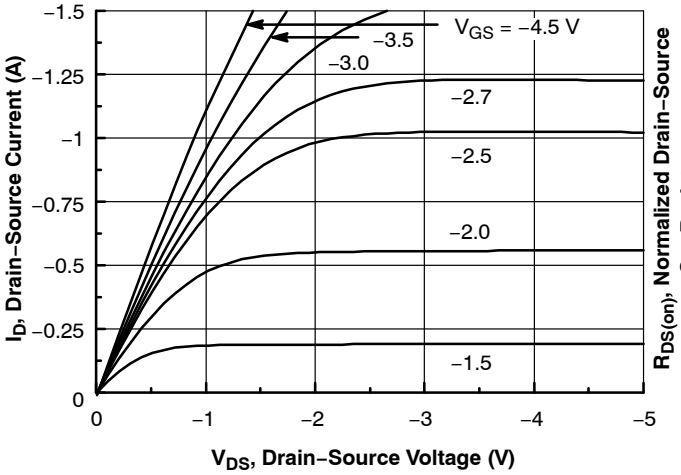


Figure 1. On−Region Characteristics

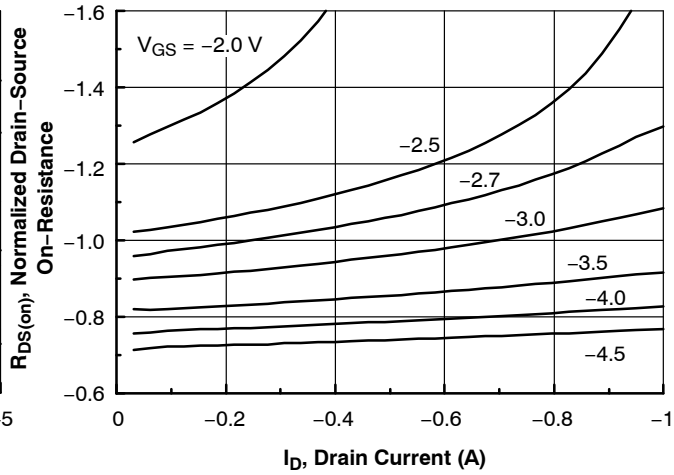


Figure 2. On−Resistance Variation with Drain Current and Gate Voltage

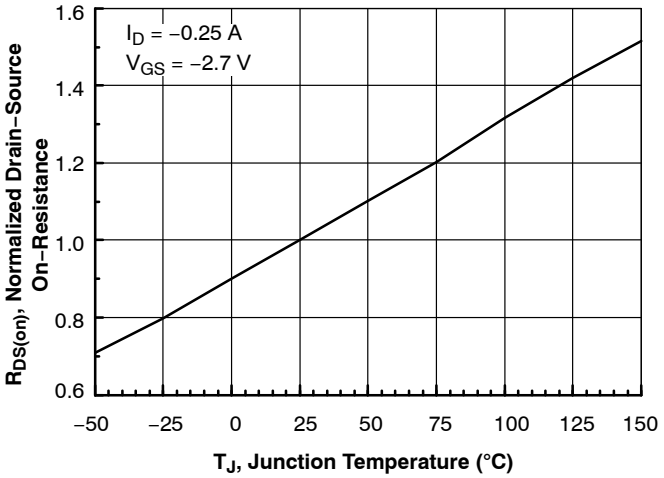


Figure 3. On−Resistance Variation with Temperature

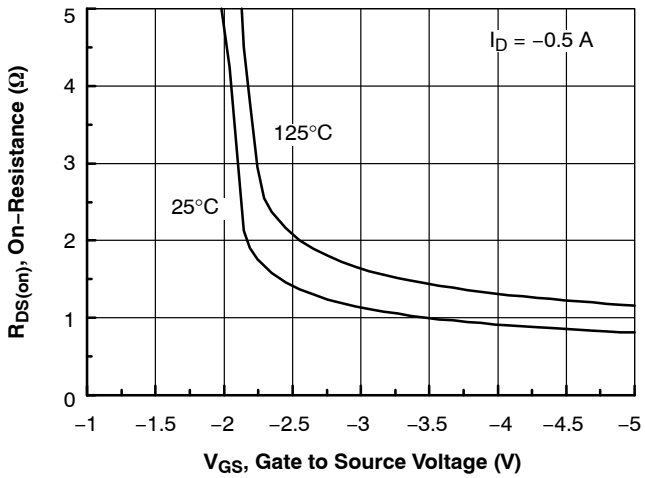


Figure 4. On Resistance Variation with Gate−To−Source Voltage

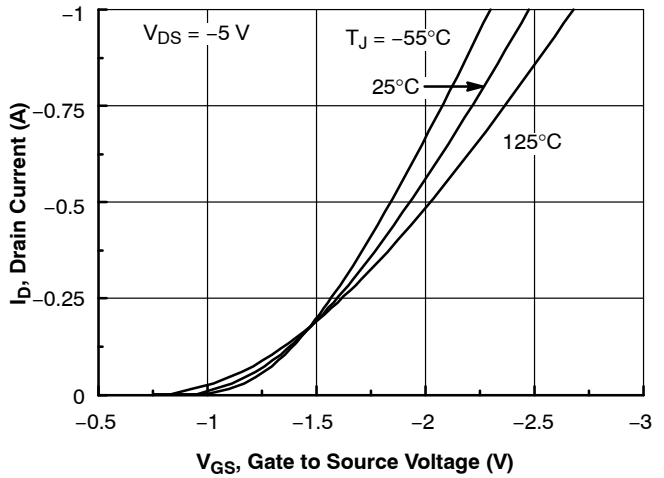


Figure 5. Transfer Characteristics

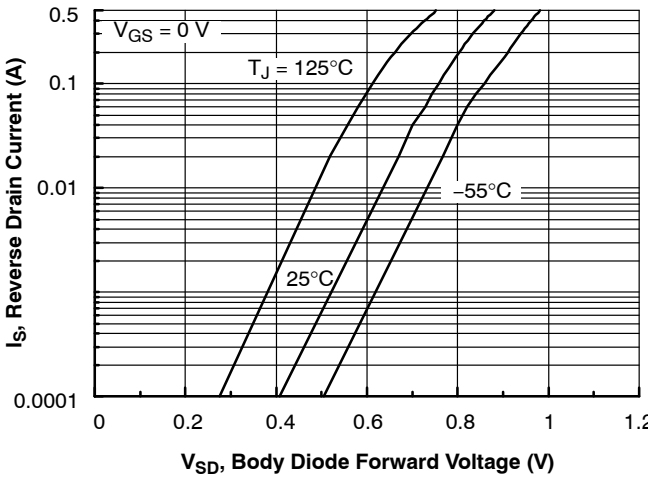


Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (continued)

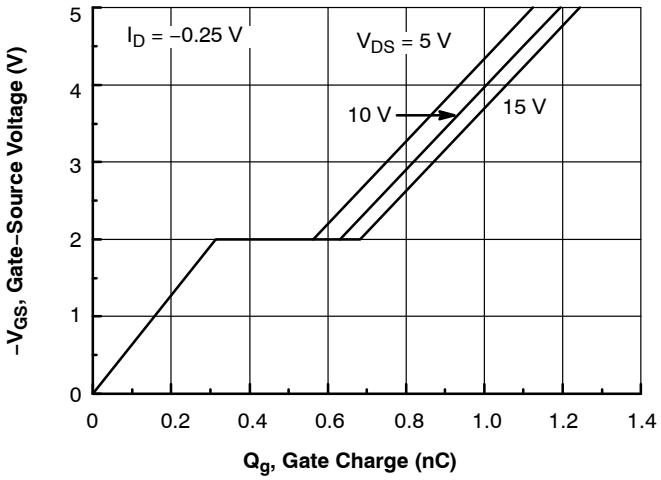


Figure 7. Gate Charge Characteristics

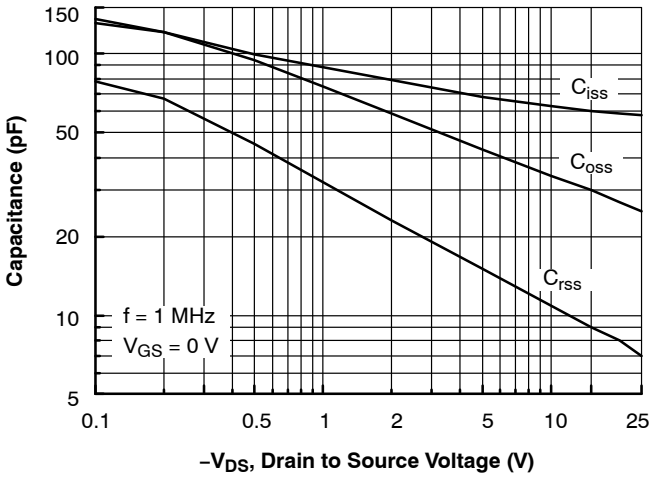


Figure 8. Capacitance Characteristics

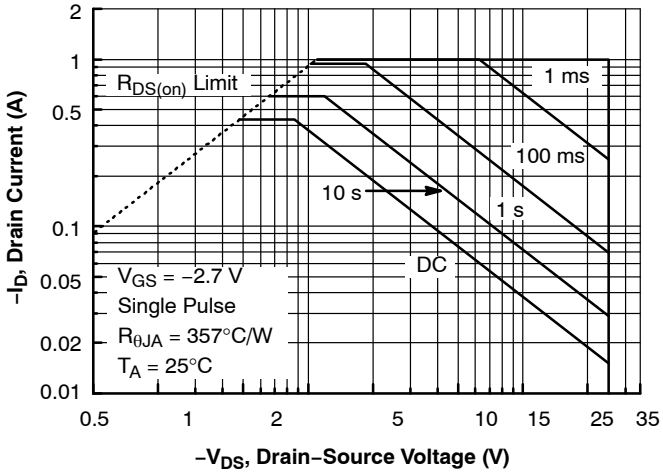


Figure 9. Maximum Safe Operating Area

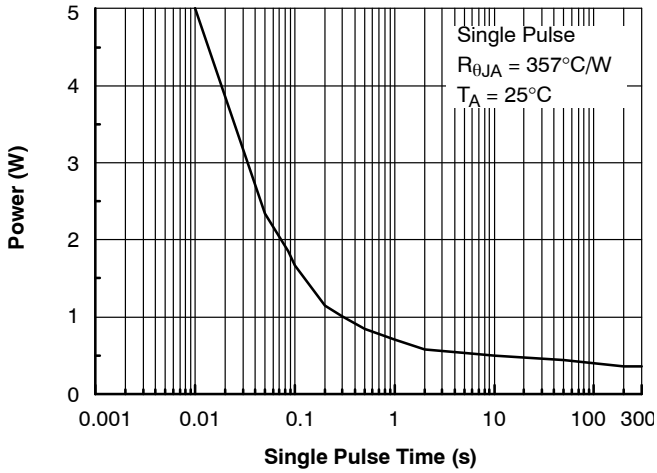


Figure 10. Maximum Pulse Maximum Power Dissipation

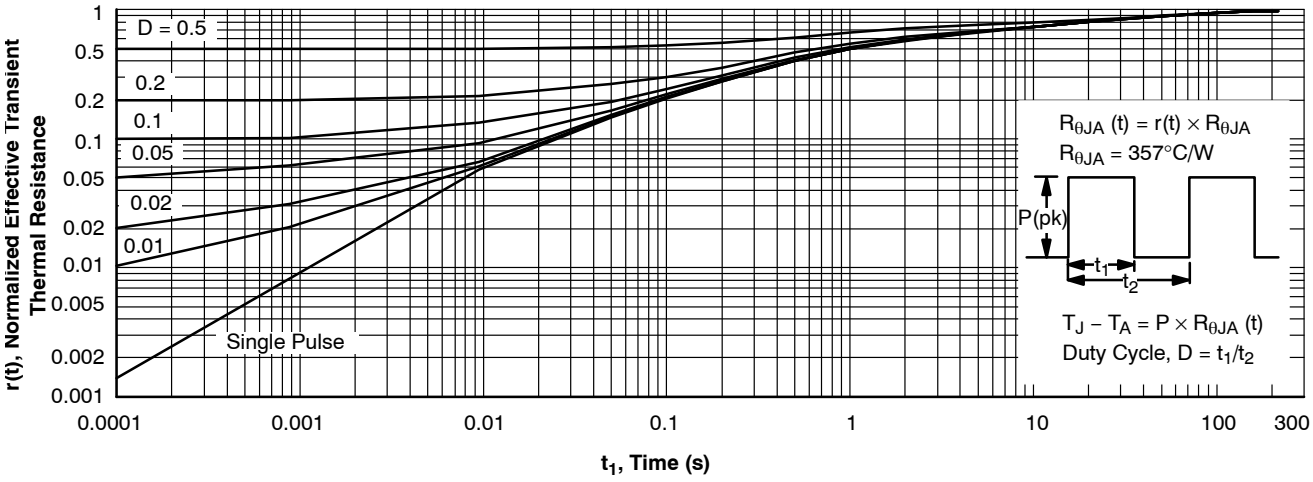
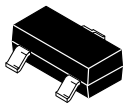


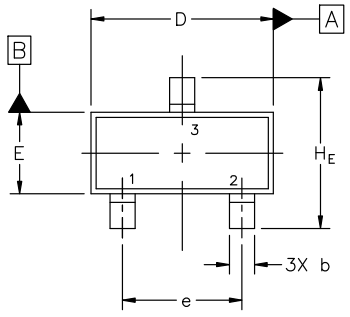
Figure 11. Transient Thermal Response Curve



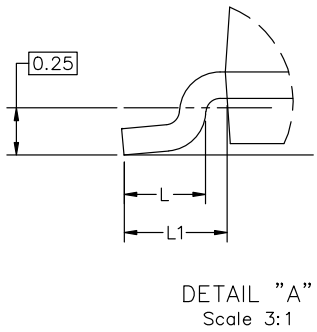
SCALE 4:1

SOT−23 (TO−236) 2.90x1.30x1.00 1.90P
CASE 318
ISSUE AU

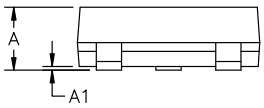
DATE 14 AUG 2024



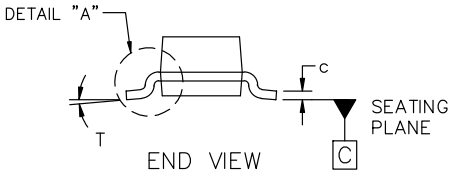
TOP VIEW



DETAIL "A"
Scale 3:1



SIDE VIEW

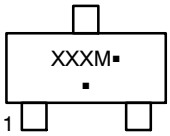


END VIEW

MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.89	1.00	1.11
A1	0.01	0.06	0.10
b	0.37	0.44	0.50
c	0.08	0.14	0.20
D	2.80	2.90	3.04
E	1.20	1.30	1.40
e	1.78	1.90	2.04
L	0.30	0.43	0.55
L1	0.35	0.54	0.69
H_E	2.10	2.40	2.64
T	0°	−−−	10°

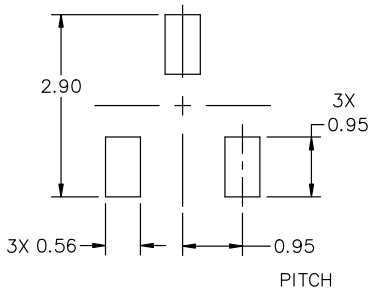
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSIONS: MILLIMETERS.
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF THE BASE MATERIAL.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.

GENERIC
MARKING DIAGRAM*



XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED
MOUNTING FOOTPRINT

* For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

STYLES ON PAGE 2

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CASE 318
ISSUE AU

DATE 14 AUG 2024

STYLE 1 THRU 5:
CANCELLED

STYLE 6:
PIN 1. BASE
2. EMITTER
3. COLLECTOR

STYLE 7:
PIN 1. EMITTER
2. BASE
3. COLLECTOR

STYLE 8:
PIN 1. ANODE
2. NO CONNECTION
3. CATHODE

STYLE 9:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 10:
PIN 1. DRAIN
2. SOURCE
3. GATE

STYLE 11:
PIN 1. ANODE
2. CATHODE
3. CATHODE−ANODE

STYLE 12:
PIN 1. CATHODE
2. CATHODE
3. ANODE

STYLE 13:
PIN 1. SOURCE
2. DRAIN
3. GATE

STYLE 14:
PIN 1. CATHODE
2. GATE
3. ANODE

STYLE 15:
PIN 1. GATE
2. CATHODE
3. ANODE

STYLE 16:
PIN 1. ANODE
2. CATHODE
3. CATHODE

STYLE 17:
PIN 1. NO CONNECTION
2. ANODE
3. CATHODE

STYLE 18:
PIN 1. NO CONNECTION
2. CATHODE
3. ANODE

STYLE 19:
PIN 1. CATHODE
2. ANODE
3. CATHODE−ANODE

STYLE 20:
PIN 1. CATHODE
2. ANODE
3. GATE

STYLE 21:
PIN 1. GATE
2. SOURCE
3. DRAIN

STYLE 22:
PIN 1. RETURN
2. OUTPUT
3. INPUT

STYLE 23:
PIN 1. ANODE
2. ANODE
3. CATHODE

STYLE 24:
PIN 1. GATE
2. DRAIN
3. SOURCE

STYLE 25:
PIN 1. ANODE
2. CATHODE
3. GATE

STYLE 26:
PIN 1. CATHODE
2. ANODE
3. NO CONNECTION

STYLE 27:
PIN 1. CATHODE
2. CATHODE
3. CATHODE

STYLE 28:
PIN 1. ANODE
2. ANODE
3. ANODE

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